Review 1: how to present

6 to 8 minutes in total: diagrams 2, demo 5, one minute spare for his questions

Timeline

Minute	What	On screen
0:00 - 0:15	One-line intro: the gist	Report, Figure 3
0:15 - 2:15	Diagrams: model, data model, data flow	Report pages 8, 10, 12
2:15 - 7:00	Demo steps 1 to 10	Browser, localhost:4000
7:00 - 8:00	Hand over the report, questions	Report

Before he comes to you

1. Run **RESET_DATA.bat** so the data is exactly as in the report (prescription #1 untouched, 10 prescriptions).
2. Run **START_DEMO.bat**. It starts the postgresql-17 and MongoDB services if needed (approve the admin prompt), starts the server on port 4000 and opens `http://localhost:4000`.
3. Check the header: both dots green, user "Abhirup Banik (pharmacist)". Close every other tab; browser zoom 110 to 125%.
4. Open the report PDF in a second window at page 8 (Figure 3). Keep pages 10 and 12 ready.
5. Keep the report open on the screenshot pages as the fallback; the table at the end maps each step to its page.

1. The diagrams (2 minutes)

Show the report, not slides. Go through three figures in this order; each builds on the last.

Proposed model (Figure 3) · page 8 · 25 s

Show: Point top to bottom: Pharmacist, React screens, API boxes, then PostgreSQL box, then MongoDB box.

The point to land: Only the API touches the databases; PostgreSQL is the transactional core, MongoDB the safety layer.

Complete data model (Figure 4) · page 10 · 40 s

Show: medicines, batches (1:N), prescriptions to items, users, dispensing_events, the red trigger box, views row, then follow the green dashed line from generic_name to the MongoDB box.

The point to land: batches is the truth for stock, dispensing_events is the append-only history, and the two databases meet only at generic_name.

Data flow, what goes where (Figure 5) · page 12 · 55 s

Show: Run your finger down the rows. Stop on "Confirm dispense": purple LOCK, orange WRITE, the green arrow to MongoDB, red BLOCK.

The point to land: Reads lock nothing; only Confirm locks, checks MongoDB again and writes stock and audit row in one transaction.

Then say "let me show it working" and switch to the browser. Do not go through Figure 9 unless he asks; the demo shows the same thing live.

2. The demo route (5 minutes)

#	Feature	Click	Point at	Time
1	Create a prescription	Dispense tab, "+ New prescription". Type Kavya Rao, 52, Dr. Arjun Rao. Warfarin 5 mg, 10. "+ Add medicine", Fluconazole 150 mg, 1. Save prescription.	Green dots; the "One transaction" note	30 s
2	Dispense with a warning	Click "Dispense 1" on Fluconazole: warning appears. Then choose prescription #1 Ramesh Iyer, click "Dispense 10" on Aspirin. Tick "I have reviewed this warning". Confirm Dispense.	Struck-through expired batch, the star, the disabled Confirm, then the trace and 165 to 155	60 s
3	Unsafe dispense refused	Choose prescription #4 Suresh Kumar, click "Dispense 4" on Sildenafil.	Red contraindicated card, no checkbox	20 s
4	Request larger than one batch	Choose "Walk-in", search Montelukast, click it, set quantity to 32. Confirm Dispense.	"takes 22", "then takes 10"; two audit events	30 s
5	Receive stock, inventory, audit log	Inventory tab, click Paracetamol 650 mg. Batch number RCV2611P, expiry 31-03-2028, quantity 120, Receive. Then Audit log tab, set the type filter to RECEIPT.	Green message; the RECEIPT row at the top	40 s
6	MongoDB flexibility and its floor	Interactions tab. Insert document (the sample is pre-filled). Then replace the text with a document without severity and Insert again.	Green "No ALTER TABLE" message; red validator message	30 s
7	Two pharmacists, the last unit	Under the hood tab, "Run the race". Wait about 2 seconds.	B's "lock acquired" time vs A's COMMIT time	40 s
8	MongoDB unreachable	"Cut MongoDB", wait for the red dot, "Attempt the dispense". Then "Restore MongoDB", wait for green, attempt again.	1533 ms ROLLBACK; then COMMIT	30 s
9	Why the FEFO query stays fast	"Run EXPLAIN ANALYZE".	Index Scan vs Seq Scan, "195x faster"	20 s
10	The audit log cannot be rewritten	"Try to edit the audit log".	The two red refusals: 42501 and append-only	20 s

If time is short, cut in this order

- MUST: step 2 (dispense with warning), step 7 (race), step 8 (MongoDB down) - these are the database concepts.
- THEN: step 1 (insert), step 5 (receive + audit log), step 10 (audit log protected).

- IF TIME: step 4 (batch spanning), step 6 (MongoDB documents), step 9 (EXPLAIN), step 3 (contraindicated).

If something goes wrong

Problem	Do this
A dot in the header is red	Run START_DEMO.bat again (it starts the services). If MongoDB stays red, show Figures 31 and 32 instead.
Page does not load / port 4000 busy	STOP_DEMO.bat, then START_DEMO.bat.
Data looks wrong (Aspirin already dispensed, extra prescriptions)	RESET_DATA.bat (about 2 s) and reload the page. Safe to run with the server running.
Prescription #1 Aspirin already shows "done"	RESET_DATA.bat. With no time for that, dispense the Warfarin line, which raises the same Aspirin + Warfarin warning.
Race button shows an error	Click it once more; it resets the Cetirizine batch itself. Otherwise Figure 30.
Running out of time	Skip to the cut list below; say "the remaining steps are in Section 4 of the report" and show the figure pages.

Screenshot fallback: report pages

Demo step	Figures	Report page
1. Create a prescription	12, 13	29
2. Dispense with a warning	14, 15, 16, 17	30, 31, 32
3. Unsafe dispense refused	19	33
4. Request larger than one batch	20, 21	33, 34
5. Receive stock, inventory, audit log	22, 23, 25	34, 35, 37
6. MongoDB flexibility and its floor	28, 29	40, 41
7. Two pharmacists, the last unit	30	42
8. MongoDB unreachable	31, 32	43
9. Why the FEFO query stays fast	33	44
10. The audit log cannot be rewritten	34	45

Rules of thumb

- Say what the database is doing, not what the screen looks like: "row lock", "one transaction", "rolled back", "partial index", "validator".
- Let the trace and the timings speak; point at the numbers.
- If he asks something you do not know, say what you do know and offer the section of the report.
- Afterwards run STOP_DEMO.bat and RESET_DATA.bat.